

1.1. Computer System

1.1.1 Definition and Features of Computer System

What is a Computer System?

A **computer system** is an electronic device that accepts data, processes it according to instructions, stores it, and produces meaningful information as output.

A complete computer system includes:

- Hardware
- Software
- Data
- Users
- Procedures

Simple Definition

A computer system is a combination of hardware and software that works together to perform tasks.

1. Hardware

The physical parts of a computer that can be touched is called Hardware.

Examples:

- Keyboard
- Mouse
- Monitor
- CPU
- Printer

2. Software

Programs and instructions that tell the computer what to do is Software.

Examples:

- Windows
- MS Word
- Google Chrome
- Games

3. Data

Data are raw facts and figures entered into the computer.

Examples:

- Numbers
- Names
- Marks
- Images

4. Information

Information is a processed and meaningful data.

Example:

- Student marks sheet
- Exam result
- Report card

5. User

A person who operates the computer is a user.

Examples:

- Student
- Teacher
- Banker
- Doctor

Features (Characteristics) of Computer System

A computer has many special features that make it useful.

1. Speed

Computers work very fast and can perform millions of calculations in a second.

Example:

- Calculating exam results within seconds.

2. Accuracy

Computers provide accurate results if the correct data and instructions are given.

Example:

- Bank calculations.

Wrong input gives wrong output. This is called **GIGO (Garbage In, Garbage Out)**.

3. Storage Capacity

Computers can store a huge amount of data and information.

Example:

- Thousands of photos, videos, and documents.

4. Diligence

A computer can work continuously for a long time without getting tired.

Example:

- Working in banks and hospitals 24 hours.

5. Automation

Computers can perform tasks automatically after instructions are provided.

Example:

- Automatic bill generation.

6. Versatility

A computer can perform many different types of tasks.

Examples:

- Playing games
- Typing documents
- Watching videos
- Coding
- Calculations

7. Reliability

Computers provide consistent and dependable results.

Example:

- ATM transactions.

8. Multitasking

A computer can perform multiple tasks at the same time.

Example:

- Listening to music while typing a document.

Advantages of Computer System

- Saves time
- Reduces human effort
- Stores large data
- Helps in communication
- Useful in education, business, and science

Disadvantages of Computer System

- Security risks (virus, hacking)
- Health problems from excessive use
- Job displacement in some fields
- Requires electricity

1.1.2 Application Areas of Computer Systems

Computers are used in almost every field of life.

1. Education

Computers are used in:

- Online classes
- Preparing notes
- Result processing
- Digital learning

Examples:

- Smart classrooms
- E-learning websites

2. Business

Businesses use computers for:

- Accounting
- Billing
- Record keeping
- Online shopping

Examples:

- Supermarkets
- E-commerce stores

3. Banking

Banks use computers for:

- ATM services
- Online banking
- Money transfer
- Account management

4. Healthcare

Hospitals use computers to:

- Store patient records
- Perform medical tests
- Diagnose diseases

Examples:

- MRI scan
- Digital reports

5. Communication

Computers help people communicate globally.

Examples:

- Email
- Video calls
- Social media
- Messaging apps

6. Entertainment

Computers are used for:

- Gaming
- Watching movies
- Music
- Animation

7. Science and Research

Scientists use computers for:

- Space research
- Weather forecasting
- Simulations
- Data analysis

8. Transportation

Used in:

- Air ticket booking
- Railway systems
- GPS navigation
- Traffic control

9. Government Offices

Computers help in:

- Record management
- National ID systems
- Tax calculations
- Census data

10. Industry and Manufacturing

Used for:

- Controlling machines
- Product design
- Automation

Example:

- Robotics in factories

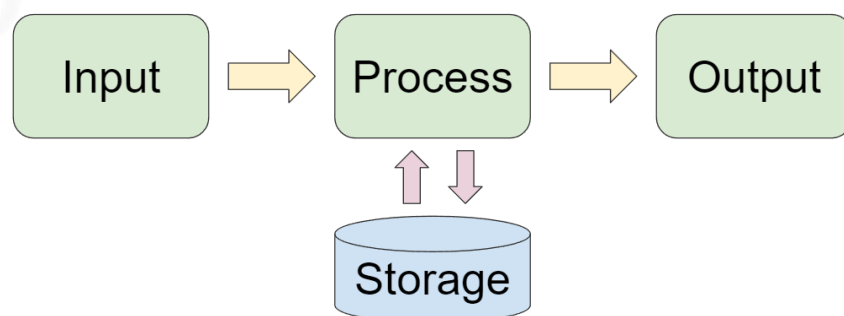
1.1.3 Functions of Computer System (IPOS Cycle)

A computer system mainly performs four basic functions known as the **IPOS Cycle**.

IPOS stands for:

1. Input
2. Processing
3. Output
4. Storage

IPOS Cycle Diagram



1. Input

Input is the process of entering data and instructions into the computer.

Input Devices

- Keyboard
- Mouse
- Scanner
- Microphone

Example

Typing your name using a keyboard.

2. Processing

Processing is the operation where the computer processes the input data according to instructions.

This work is done by the:

- CPU (Central Processing Unit)

Example

Calculating total marks.

3. Output

Output is the final meaningful result produced by the computer.

Output Devices

- Monitor
- Printer
- Speaker

Example

Displaying exam results on screen.

4. Storage

Storage means saving data and information for future use.

Storage Devices

- Hard Disk
- SSD
- Pen Drive
- Memory Card

Example

Saving homework files.

1.2 Input Devices

Introduction to Input Devices

Input devices are the hardware components used to enter data and instructions into a computer system. They help users communicate with the computer. Without input devices, a computer cannot receive information from users.

Definition of Input Device

An **input device** is a hardware device used to send data, commands, or instructions to a computer.

Common Input Devices

According to the syllabus, the main input devices are:

1. Mouse
2. Keyboard
3. Joystick
4. Microphone
5. Scanner
6. Webcam
7. Touchscreen
8. Touchpad

1. Mouse

A mouse is a pointing input device used to control the movement of the cursor on the screen.

Main Parts of a Mouse

- Left button
- Right button
- Scroll wheel
- Optical sensor

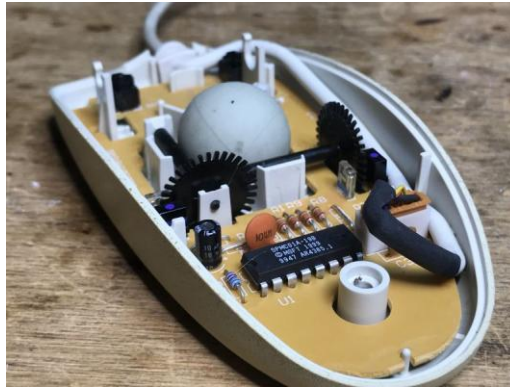
Functions of Mouse

- Pointing
- Clicking
- Double-clicking
- Dragging
- Scrolling

Types of Mouse

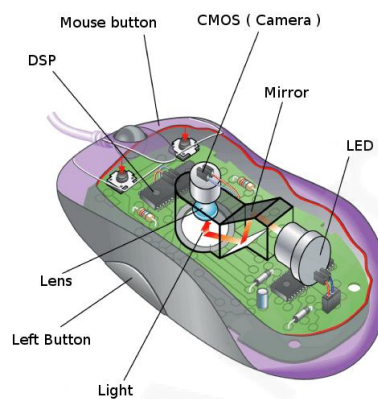
1. Mechanical Mouse

Uses a rubber ball for movement.



2. Optical Mouse

Uses light sensors for movement.



3. Wireless Mouse

Connects through Bluetooth or USB receiver.



Advantages

- Easy to use
- Fast navigation
- Useful for drawing and gaming

Disadvantages

- Requires flat surface
- Can be damaged easily

2. Keyboard

A keyboard is an input device used to type text, numbers, and commands into a computer.

Main Sections of Keyboard

1. Alphabet Keys

Used to type letters.

2. Numeric Keys

Used to type numbers.

3. Function Keys (F1–F12)

Used for special functions.

4. Arrow Keys

Used for navigation.

5. Special Keys

Examples:

- Enter
- Shift
- Ctrl
- Alt
- Backspace

Functions of Keyboard

- Typing documents
- Entering commands
- Shortcut operations

Advantages

- Fast typing
- Easy text input

Disadvantages

- Requires typing skills
- Long use may cause finger strain



3. Joystick

A joystick is an input device used mainly for controlling movement in games and simulations.

Uses of Joystick

- Video games
- Flight simulators
- Robotics control

Advantages

- Better control in games
- Smooth movement control

Disadvantages

- Limited use outside gaming
- Difficult for beginners



4. Microphone

A microphone is an input device used to enter sound or voice into the computer.

Uses

- Voice recording
- Video calls
- Online classes
- Voice commands

Advantages

- Easy voice communication
- Useful for recording

Disadvantages

- Background noise affects quality
- Requires proper setup

5. Scanner

A scanner is an input device that converts paper documents or images into digital form.

Types of Scanner

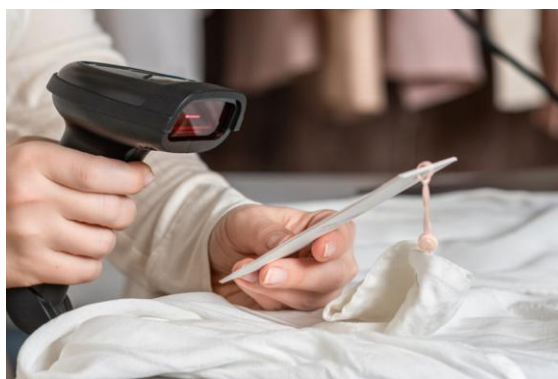
1. Flatbed Scanner

Most common type.



2. Handheld Scanner

Portable scanner.



3. Barcode Scanner

Used in shops and supermarkets.

Uses

- Scanning documents
- Digitizing photos
- Record keeping

Advantages

- Preserves documents digitally
- High-quality image capture

Disadvantages

- Can be slow
- Large files require storage



6. Webcam

A webcam is a digital camera connected to a computer used for capturing live images and videos.

Uses

- Video conferencing
- Online meetings
- Live streaming
- Online classes

Advantages

- Real-time communication
- Easy online interaction

Disadvantages

- Privacy concerns
- Requires internet for communication

7. Touchscreen

A touchscreen is both an input and output device that allows users to interact by touching the screen.

Uses

- Smartphones
- ATMs
- Tablets
- Information kiosks

Advantages

- Easy to use
- Fast interaction

Disadvantages

- Fingerprints on screen
- Expensive to repair

8. Touchpad

A touchpad is a touch-sensitive surface mainly found in laptops used to control the cursor.

Functions

- Moving cursor
- Clicking
- Scrolling
- Zooming

Advantages

- Built into laptops
- No extra space needed

Disadvantages

- Less precise than mouse
- Difficult for gaming

Comparison of Input Devices

Device	Main Function	Common Use
Mouse	Cursor control	Navigation
Keyboard	Typing	Text input
Joystick	Direction control	Gaming
Microphone	Audio input	Recording
Scanner	Image/document input	Digitization
Webcam	Video input	Video calls
Touchscreen	Touch interaction	Mobile devices
Touchpad	Cursor control	Laptops

1.4 Motherboard and Data Bus

Introduction to Motherboard

The motherboard is one of the most important parts of a computer system. It is the **main circuit board** that connects all hardware components together so they can communicate with each other. Without the motherboard, different parts of the computer cannot work together.

1.4.1 Role of Motherboard in Connecting Components

Definition of Motherboard

A **motherboard** is the main printed circuit board (PCB) of a computer that holds and connects important hardware components.

It is also called:

- Mainboard
- System board

Components Connected to Motherboard

The motherboard connects:

- CPU
- RAM
- Hard Disk
- SSD
- Keyboard
- Mouse
- Graphics Card
- Sound Card
- Power Supply
- Ports and connectors

Main Functions of Motherboard

1. Connecting Hardware Components

It connects all parts of the computer together.

2. Communication Between Devices

It allows devices to communicate with each other.

Example:

- CPU communicating with RAM.

3. Power Distribution

It distributes electricity from the power supply to different components.

4. Supporting Expansion

It allows additional hardware to be added.

Examples:

- Graphics card
- Network card

5. Housing Important Circuits

The motherboard contains important electronic circuits and chips.

Important Parts of Motherboard

Part	Function
CPU Socket	Holds the CPU
RAM Slot	Holds memory modules
Expansion Slots	Add extra cards
BIOS/UEFI Chip	Starts the computer
Ports	Connect external devices
Chipset	Controls data flow

Features of Motherboard

- Large circuit board
- Contains electronic pathways
- Supports hardware communication
- Provides ports and slots

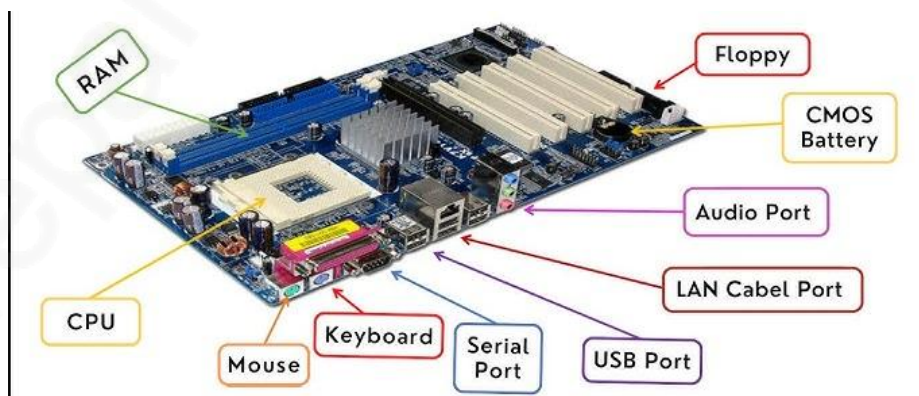
Advantages of Motherboard

- Makes communication possible
- Organizes all components
- Supports upgrades

Disadvantages

- Damage can stop the whole computer
- Expensive to replace

Simple Diagram of Motherboard



1.4.2 Importance of Data Bus in Data Transfer

What is a Data Bus?

A **data bus** is a communication pathway that transfers data between computer components.

It acts like a road for data movement.

Function of Data Bus

The data bus transfers:

- Data
- Instructions
- Information

between:

- CPU
- Memory
- Input devices
- Output devices

Example

When you type using a keyboard:

1. Data travels through the data bus.
2. CPU processes it.
3. Output appears on the monitor.

Types of Bus in Computer

1. Data Bus

Transfers actual data.

2. Address Bus

Transfers memory addresses.

3. Control Bus

Transfers control signals.

Importance of Data Bus

1. Fast Data Transfer

Helps transfer data quickly.

2. Communication

Allows devices to communicate.

3. System Performance

A faster bus improves computer performance.

Bus Width

The amount of data transferred at once is called **bus width**.

Examples:

- 8-bit
- 16-bit
- 32-bit
- 64-bit

A wider bus transfers more data at one time.

Comparison of Bus Types

Bus Type	Function
Data Bus	Transfers data
Address Bus	Transfers addresses
Control Bus	Transfers signals

1.5 Memory (Primary and Secondary)

Computer memory is the storage area where data and instructions are stored for processing and future use.

Functions of Memory

1. Stores Data

Keeps raw data entered by users.

2. Stores Instructions

Stores programs and commands.

3. Stores Results

Stores processed information.

4. Helps Fast Processing

Allows CPU to access data quickly.

Types of Memory

Memory is mainly divided into:

1. Primary Memory
2. Secondary Memory

Primary Memory

Primary memory is directly accessed by the CPU.

It is also called:

- Main Memory
- Internal Memory

Characteristics of Primary Memory

- Fast speed
- Smaller storage
- Expensive
- Directly connected to CPU

Examples

- RAM
- ROM
- Cache
- Register

Secondary Memory

Secondary memory stores data permanently.

It is also called:

- External Memory
- Auxiliary Memory

Characteristics

- Large storage
- Slower than primary memory
- Cheaper
- Permanent storage

Examples

- HDD
- SSD
- CD/DVD
- Pen drive

1.5.2 Units of Memory Measurement

Memory is measured using units.

Basic Units

Unit	Meaning
Bit	Smallest unit
Nibble	4 bits
Byte	8 bits

Larger Units

Unit	Value
1 KB	1024 Bytes

1 MB	1024 KB
1 GB	1024 MB
1 TB	1024 GB

Example

- A photo may take several MB.
- A movie may take several GB.

1.5.3 Types of Memory and Comparison

Primary Memory	Secondary Memory
Faster	Slower
Temporary	Permanent
Smaller capacity	Larger capacity
Expensive	Cheaper
Directly accessed by CPU	Indirectly accessed

1.5.4 RAM, ROM, Cache, Register

1. RAM (Random Access Memory)

RAM is temporary memory used while the computer is running.

It is called:

- Volatile memory

because data is lost when power is turned off.

Functions

- Stores currently running programs
- Helps fast processing

Characteristics

- Fast
- Temporary
- Read and write memory

Example

Opening a game or browser uses RAM.



2. ROM (Read Only Memory)

ROM stores permanent instructions needed to start the computer.

It is:

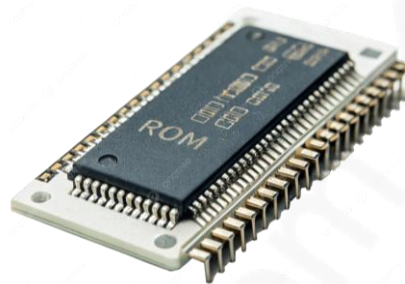
- Non-volatile memory

Functions

- Stores booting instructions

Characteristics

- Permanent
- Cannot be easily changed



3. Cache Memory

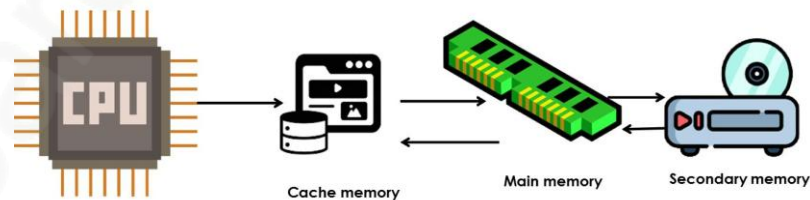
Cache memory is very high-speed memory located close to the CPU.

Function

Stores frequently used data for faster access.

Characteristics

- Very fast
- Small storage
- Expensive



4. Register

Registers are very small memory units inside the CPU.

Function

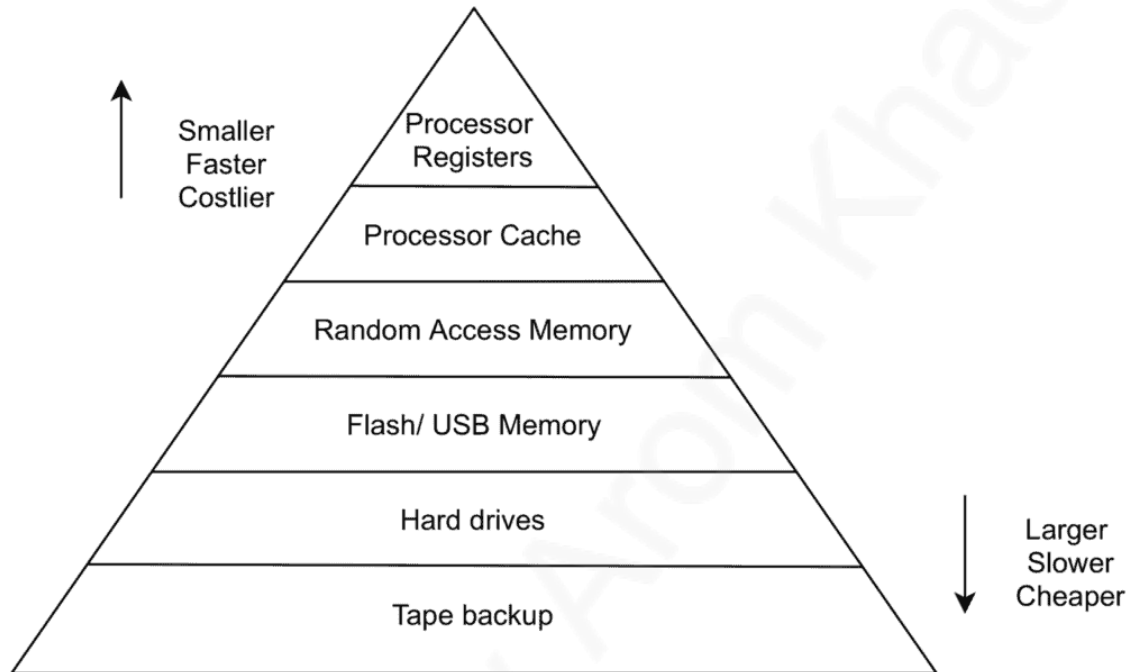
Temporarily store data during processing.

Characteristics

- Fastest memory
- Very small capacity

Comparison of RAM and ROM

RAM	ROM
Temporary	Permanent
Volatile	Non-volatile
Read and write	Mostly read only
Faster	Slower



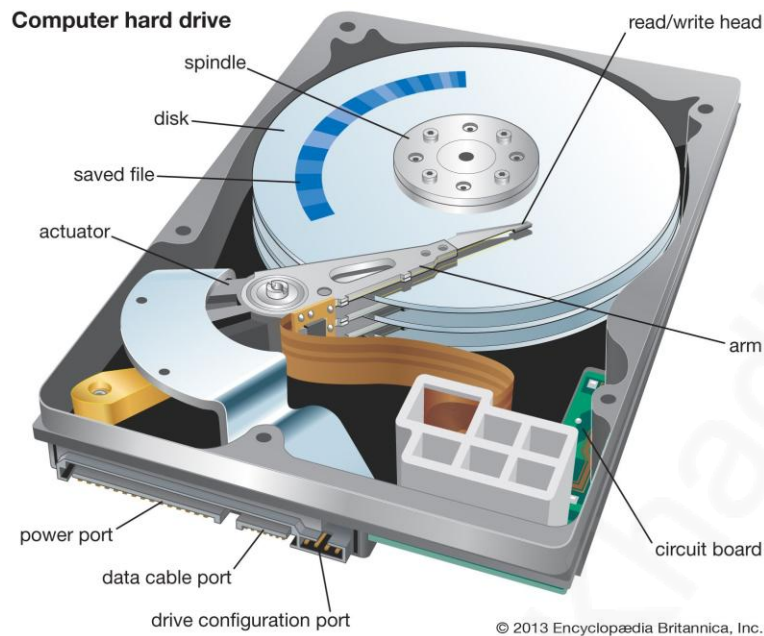
1.5.5 HDD, SSD, CD/DVD, Flash Memory

1. HDD (Hard Disk Drive)

A storage device that uses magnetic disks to store data.

Characteristics

- Large storage
- Cheaper
- Slower than SSD



2. SSD (Solid State Drive)

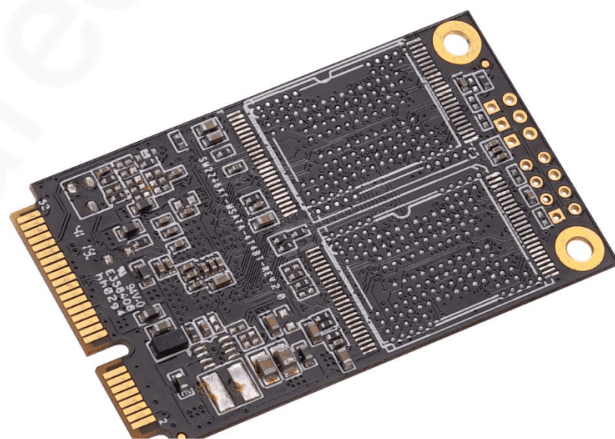
A storage device that uses flash memory.

Characteristics

- Faster
- Durable
- Expensive

HDD vs SSD

HDD	SSD
Mechanical parts	No moving parts
Slower	Faster
Cheaper	More expensive
Noisy	Silent



3. CD/DVD

Optical storage devices used to store data using laser technology.

Uses

- Movies
- Software installation
- Music storage

CD	DVD	Blue-ray
		
650 - 700 MB of data	Single-layer: 4.7 GB Dual-layer: 8.5 GB	Single-layer: 25 GB Dual-layer: 50 GB Quaruple-layer: 120 GB

4. Flash Memory

Electronic memory used in:

- Pen drives
- Memory cards

Characteristics

- Portable
- Rewritable
- Durable



1.5.6 Functions and Characteristics of Memory

Characteristics of Good Memory

1. Speed

How fast data can be accessed.

2. Capacity

Amount of data that can be stored.

3. Reliability

Ability to store data safely.

4. Cost

Price of memory devices.

Functions of Memory

- Stores data
- Stores instructions
- Stores output
- Helps CPU processing

1.6 Output Devices

An **output device** is a hardware device that receives information from a computer and presents it to the user in a readable form.

Examples of Output Devices

- Monitor
- Printer
- Speaker
- Projector
- Headphones

Functions of Output Devices

1. Display processed information
2. Produce hard copy or soft copy
3. Help users understand results
4. Provide audio and visual output

Types of Output

1. Soft Copy Output

Output displayed electronically.

Examples:

- Monitor display
- Audio from speaker

2. Hard Copy Output

Output printed on paper.

Examples:

- Printed documents
- Photos

1.6.1 Overview of Output Devices

Output devices convert digital information into human-understandable form.

They mainly provide:

- Visual output
- Audio output
- Printed output

Common Output Devices

Device	Type of Output
Monitor	Visual
Printer	Printed
Speaker	Audio
Projector	Visual
Plotter	Graphic printing

1.6.2 LED and LCD Monitor

Monitor

A monitor is an output device that displays information visually on the screen.

It is also called:

- Visual Display Unit (VDU)

Types of Monitor

1. LCD Monitor

Full Form

Liquid Crystal Display

Features

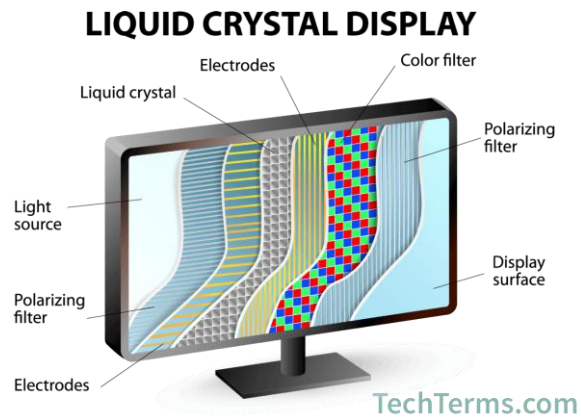
- Thin and lightweight
- Uses liquid crystals
- Consumes less electricity

Advantages

- Better picture quality
- Energy efficient
- Less heat production

Disadvantages

- More expensive than older CRT monitors
- Limited viewing angle



2. LED Monitor

Full Form

Light Emitting Diode

Features

- Uses LED backlight
- Brighter display
- Better color quality

Advantages

- Energy efficient
- Thin design
- High brightness
- Better contrast

Disadvantages

- Costlier than LCD



Difference Between LCD and LED

LCD	LED
Uses fluorescent backlight	Uses LED backlight
Lower brightness	Higher brightness

Slightly thicker	Thinner
Less energy efficient	More energy efficient

1.6.3 Laser, Inkjet, Dot Matrix and 3D Printer

Printer

A printer is an output device that produces hard copy output on paper.

Types of Printers

Printers are mainly divided into:

1. Impact Printers
2. Non-Impact Printers

1. Dot Matrix Printer

A dot matrix printer is an impact printer that prints characters using small pins striking an ink ribbon.

Features

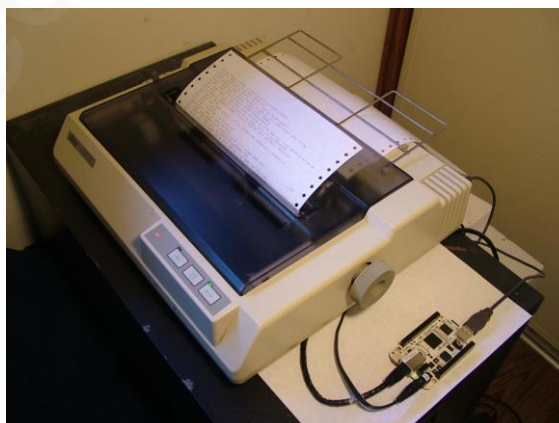
- Noisy
- Slow printing
- Can print carbon copies

Advantages

- Cheap printing cost
- Durable

Disadvantages

- Low print quality
- Noisy operation



2. Inkjet Printer

An inkjet printer sprays tiny droplets of ink onto paper.

Features

- Good color printing
- Common for home use

Advantages

- High-quality color printing
- Affordable

Disadvantages

- Ink can be expensive
- Slower than laser printers



3. Laser Printer

A laser printer uses laser technology and toner powder for printing.

Features

- Fast printing
- High-quality output

Advantages

- Very fast
- Clear printing
- Suitable for offices

Disadvantages

- Expensive
- Higher initial cost



4. 3D Printer

A 3D printer creates three-dimensional physical objects layer by layer.

Uses

- Engineering
- Medical field
- Product design
- Prototyping

Advantages

- Produces real objects
- Useful for design testing

Disadvantages

- Expensive
- Slow production



Comparison of Printers

Printer	Speed	Quality	Noise
Dot Matrix	Slow	Low	High
Inkjet	Medium	Good	Low
Laser	Fast	Excellent	Low

1.6.4 USB/Type C, HDMI, VGA, Ethernet, Audio Ports

Computer Ports

Ports are connection points used to connect external devices to a computer.

1. USB Port

Full Form

Universal Serial Bus

Functions

- Connect keyboard
- Connect mouse
- Transfer data
- Charge devices

Features

- Fast data transfer
- Easy connection

2. Type-C Port

A modern USB connector that supports:

- Fast charging
- Fast data transfer
- Video transmission

Advantages

- Reversible connector
- Faster performance

3. HDMI Port

Full Form

High Definition Multimedia Interface

Functions

Transfers:

- High-quality video
- Audio

to monitors, TVs, and projectors.

Advantages

- Single cable for audio and video
- High-quality transmission

4. VGA Port

Full Form

Video Graphics Array

Functions

Used for video signal transmission to monitors and projectors.

Characteristics

- Older technology
- Supports only video

Difference Between HDMI and VGA

HDMI	VGA
Transfers audio and video	Transfers only video
Digital signal	Analog signal
Better quality	Lower quality

5. Ethernet Port

A port used to connect a computer to a wired network or internet.

Uses

- Internet connection
- Networking

Features

- Stable connection
- Fast network speed

6. Audio Ports

Ports used for audio input and output.

Types

- Headphone port
- Microphone port
- Speaker port

Importance of Ports

- Connect external devices
- Allow communication
- Support multimedia functions
- Enable networking



1.7 Computer Software

Introduction to Software

Software is an important part of a computer system. Without software, hardware cannot perform any task. Software gives instructions to the computer.

Definition of Software

Software is a collection of programs and instructions that tell the computer what to do.

1.7.1 Concept and Types of Software

Types of Software

Software is mainly divided into:

1. System Software
2. Application Software

1. System Software

System software controls and manages computer hardware and provides a platform for application software.

Examples

- Operating System
- Device Drivers
- Utility Programs

Operating System (OS)

An operating system manages all hardware and software resources.

Examples

- Windows
- Linux
- Android
- macOS

Functions of Operating System

- Manages files
- Controls devices
- Runs applications
- Provides user interface

Utility Software

Utility software helps maintain and protect the computer.

Examples

- Antivirus
- Disk Cleaner
- Backup software

Device Driver

A device driver helps the operating system communicate with hardware devices.

Example

Printer driver.

2. Application Software

Application software is designed to perform specific user tasks.

Examples

- MS Word
- Excel
- VLC Media Player
- Games

Types of Application Software

1. General Purpose Software

Used for common tasks.

Examples:

- Word processors
- Browsers

2. Customized Software

Developed for specific organizations.

Examples:

- School management system
- Hospital software

Difference Between System and Application Software

System Software	Application Software
Runs computer system	Performs specific tasks
Essential for operation	Optional for users
Starts before applications	Runs after OS starts

1.7.2 Open Source and Proprietary Software

Open Source Software

Software whose source code is freely available for use and modification.

Examples

- Linux
- LibreOffice
- Mozilla Firefox

Advantages

- Free to use
- Customizable
- Community support

Disadvantages

- Limited official support
- May require technical knowledge

Proprietary Software

Software owned by a company or person with restricted source code access.

Examples

- Microsoft Windows
- Adobe Photoshop
- MS Office

Advantages

- Professional support
- User-friendly

Disadvantages

- Expensive
- Limited customization

Difference Between Open Source and Proprietary Software

Open Source	Proprietary
Source code available	Source code hidden
Usually free	Paid license
Can modify software	Cannot modify software

1.7.3 System Software vs Application Software

System Software	Application Software
Controls computer	Helps users perform tasks
Runs in background	Runs according to user needs
Example: Windows	Example: MS Word

1.7.4 Mobile and Web Application

Mobile Application

Definition

Software designed to run on smartphones and tablets.

Examples

- WhatsApp
- TikTok
- Facebook
- Instagram

Features

- Touch-friendly
- Portable
- Internet-enabled

Web Application

Software accessed through a web browser using the internet.

Examples

- Google Docs
- Gmail
- YouTube

Features

- Runs in browser
- No installation needed
- Accessible anywhere

Difference between Mobile and Web Application

Mobile App	Web App
Installed on device	Accessed through browser

Works faster	Depends on internet
Uses device hardware	Limited hardware access

Advantages of Software

- Makes hardware useful
- Automates tasks
- Increases productivity
- Helps communication

Disadvantages of Software

- Viruses and malware
- Software piracy
- Compatibility problems